

Software engineer building real-time multimodal AI in production — Facilitator, Microsoft’s agent that participates in live, multi-person Teams meetings. Seven years shipping real-time, multimodal human-computer interaction at scale: expressive avatars, embodiment, and XR for Microsoft Mesh. M.S. in Computer Science (Human-Computer Interaction) in progress at Georgia Tech.

- CORE COMPETENCIES

- Real-Time Multimodal Interaction
 - Human-Computer Interaction Design

- Conversational AI Systems
 - Expressive Avatars & Embodiment

- Cross-Platform UI System
 - Performance Optimization

TECHNICAL SKILLS

Languages: C#, C++, TypeScript, JavaScript, Python, Swift
AI / Cloud / Tools: LLMs, Agents, RAG, MCP, A2A, Skills, Ralph Loop, PyTorch, OpenCV, Azure, ROS, React, Git
XR / Graphics / 3D: Unity, Unreal, OpenXR, WebXR, visionOS, OpenGL, Vulkan, DirectX, Three.js, Babylon.js, Blender, 3D Math

PROFESSIONAL EXPERIENCE

- Microsoft

Software Engineer II, Teams AI

Redmond, WA
September 2024 – Present

- Build Facilitator, Microsoft’s conversational AI agent that participates in live, multi-party Teams meetings — real-time notes, agenda tracking, decision capture, and in-conversation Q&A — shipping to Microsoft 365 Copilot customers.
 - Design the turn-taking layer of a proactive agent — when and how AI speaks into a live group conversation without interrupting — plus early-stage multimodal LLM initiatives (details under NDA).Driving early-stage AI initiatives integrating multimodal LLMs into spatial collaboration experiences (details under NDA).
- Software Engineer II, Mixed Reality

June 2022 – September 2024

- Engineered the shared UI framework powering Microsoft Mesh and Avatars in Teams for millions of users — defined reusable interaction patterns, input abstractions, accessibility integration, and spatial UI components adopted by every Mesh product team.
 - Owned the real-time avatar rendering and performance pipeline across HoloLens, Quest, web, and desktop — profiled CPU/GPU frame budgets and applied LOD switching, GPU instancing, and other optimizations to sustain target framerates on constrained hardware.
 - Shipped end-to-end avatar customization and character-building systems with data-driven content pipelines, weekly release cadence, and 20+ language localization
- Holos Inc.

AR/VR Interaction Engineer

<https://holos.io/>
February 2019 – May 2022

- Build an interactive, networked AR/VR training simulation system with hand tracking, gesture recognition, and simulated physics for enterprise content management.
 - Prototype and ship multiplayer networking, 3D model processing, and virtual object manipulation features—directly contributing to a \$750K U.S. Air Force research contract and TechStars acceptance.

RESEARCH

- Project Virtualso

Founder & Developer (Unity | OpenXR | C++ | C#)

Online
January 2020 – June 2023

- Built a VR interview and presentation simulator with NLP-powered conversational AI agents featuring emotion-driven facial expressions and real-time gesture feedback; validated with business-school students for interview practice.
- Kats Laboratory of Applied Physics

AR/VR Researcher

Madison, WI
October 2017 – March 2022

- Built chromatic-adjustment AI algorithms on hyperspectral imagery (90% accuracy simulating color-vision deficiency) and colorcalibrated VR simulations on Oculus HMDs; co-authored arXiv publication.
- UW HCI Group, People and Robots Lab

AR/VR Researcher

Madison, WI
April 2017 – December 2017

- Added speech recognition and NLP-driven interaction to children’s reading-companion robots; validated in an eight-family trial.

EDUCATION AND CREDENTIALS

Georgia Institute of Technology | M.S. in Computer Science (Human-Computer Interaction) Expecting Spring 2027
University of Wisconsin-Madison | B.A. in Computer Science & B.A in Communication Arts August 2016 – June 2020

- Teaching assistant for CS 559 (Computer Graphics) for 350+ students across 2 semesters